# TANMAY KOTAK

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## SUMMARY

Fourth-year Computer Engineering student specializing in backend engineering and applied AI systems. Experienced in building production APIs, enterprise integrations, RAG pipelines, and agentic AI systems across Python, TypeScript, and Node.js

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, Java, Python
**AI/ML & Cloud:** LangGraph, LangChain, RAG, LLM APIs, HuggingFace, Firebase, Vercel
**Frontend:** React.js, Next.js, React Query, TailwindCSS
**Backend:** Node.js, Express.js, Nest.js, FastAPI, REST API Design, Microservices
**Databases:** PostgreSQL, Prisma ORM, MongoDB, MySQL

## EXPERIENCE

**Software Engineering Intern** June 2026 – October 2026
Krios Info Solutions Pvt. Ltd. · Wakad, Pune

- **Spearheaded backend architecture and production rollout** of an enterprise ASN supply-chain portal, translating requirements from procurement leads and SAP teams into **130+ REST APIs**, a **28-model MySQL schema** using Prisma ORM, and 9 bi-directional SAP S/4HANA integration pipelines.
- Engineered the **inbound logistics and delivery scheduling engine** for manufacturing plants and vendor networks, supporting real-time shipment coordination, multi-batch commitments, and gate check-ins with transaction-level revision guards and multi-step state machines for safe order modifications and rollbacks.
- Built an **asynchronous document intelligence pipeline** with OCR-based extraction and purchase-order validation, alongside automated tax and valuation processing to eliminate manual gate check-in bottlenecks and prevent billing discrepancies before SAP posting.

**Freelance Web Developer** November 2025
Beauty & Bangs Live Link · Client Project

- Delivered an SSG Next.js/Tailwind production client site with perfect 100/98/100/100 Lighthouse scores, asset optimization, and zero render-blocking resources.

## PROJECTS

**DocMind – Multi-Tenant RAG Document Intelligence Platform** · FastAPI · https://github.com/TanmayKQ/DocMind
PostgreSQL (pgvector) · React.js · Gemini API

- Built a hybrid retrieval pipeline combining pgvector semantic search with PostgreSQL tsvector full-text search in a single CTE query, improving retrieval precision via 70/30 semantic-lexical weighting.
- Implemented real-time streaming via Server-Sent Events (SSE) with minimal time-to-first-token, piping Gemini 2.5 Flash tokens through an async generator to a React typing interface.
- Designed an async document ingestion pipeline with 500-token chunking and batch embeddings (20 chunks/req), returning HTTP 202 in under 1s with non-blocking background workers.
- Architected 12 REST endpoints with JWT auth, tenant isolation, and token-bucket rate limits; managed schema evolution via version-controlled Alembic migrations.

**DevAgent – Autonomous CLI Coding Assistant** · Python · LangGraph · Node.js · https://github.com/TanmayKQ/DevAgent
TypeScript

- Developed an agentic AI coding assistant with a decoupled Python/LangGraph reasoning core and Node.js/TypeScript execution harness communicating through schema-validated JSONL IPC.
- Architected a LangGraph state machine to plan coding tasks, execute filesystem operations, evaluate test results, and iteratively recover from execution failures.
- Implemented controlled code execution using repository-scoped directory isolation, shell-less subprocesses, and a two-phase plan-then-commit workflow for approving file modifications.
- Built a deterministic E2E evaluation framework across 17 automated coding benchmarks, using dependency injection to mock LLM responses for reproducible, cost-free CI testing.

**IntelStudy – AI-Powered Study Assistant** · React.js · Node.js · MongoDB · Firebase · https://github.com/TanmayKQ/IntelStudy
· HuggingFace

- Designed a document ingestion REST API (Node.js/Express.js) parsing PDFs and chunking content before dispatching to HuggingFace Inference APIs for automated summarization and quiz generation.
- Built resilient AI inference layer with exponential backoff retry, circuit-breaker pattern, and structured JSON fallbacks, guaranteeing graceful degradation during upstream model downtime.
- Secured backend using Firebase Authentication, Helmet.js, and API rate limiting; isolated environment secrets across dev and production runtimes.

## EDUCATION

**D.Y. Patil College of Engineering, Akurdi** – Pune, Maharashtra Aug 2023 – Jun 2027
B.E. in Computer Engineering | CGPA: 7.17 / 10.0
**Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Theory of Computation